

Dry cell types

1. **Carbon-zinc.** These are primary (non-rechargeable type) dry cells.
2. **Alkaline.** These dry cell batteries use the zinc-manganese dioxide chemistry. Most alkaline batteries are non-rechargeable type, although some are designed to be recharged. Compared to zinc-carbon batteries, alkaline batteries have a higher energy density and longer shelf-life, with the same voltage. These use an alkaline electrolyte of potassium hydroxide, instead of the acidic ammonium chloride or zinc chloride electrolyte of zinc-carbon batteries. Alkaline batteries are used in many household items such as MP3 players, CD players, digital cameras, pagers, toys, lights and radios, to name a few.
3. **Portable rechargeable.** These include several sealed 'dry cell' types, which are useful in applications such as mobile phones and laptop computers. Cells of this type (in the order of increasing power density and cost) include nickel-cadmium (NiCd), nickel-zinc (NiZn), nickel-metal hydride (NiMH) and lithium-ion (Li-ion) cells.

— Based on a post on Wikipedia

key customers. In the mid term we are also considering serious investments in India, to become a leading player in energy solutions by 2016," shares Shah.

DRY CELL BATTERIES

Dry cell batteries are consumed predominantly by economically weaker sections of rural India as an alternate and portable source of energy for lighting, entertainment, safety and security. In fact, the price-sensitive rural market accounts for as much as 65-70 per cent of the demand.

Size of the dry cell batteries market in India is estimated at 2.5 billion pieces by volume and ₹ 12.5 billion by value. Within the industry, carbon-zinc batteries (which are non-rechargeable) account for 97 per cent of the market, while alkaline and other varieties of batteries account for the rest, according to Subramania Kumaraswami, secretary, Association of Indian Dry Cell Manufacturers.

There are only three manufacturers in the organised sector, producing carbon-zinc dry cell batteries. The demand for alkaline, rechargeable and other varieties is met through imports.

Industry status. The battery market went through some significant volatility over the last five years. The industry experienced adverse impact of the unprecedented rise in input material costs. These adverse impacts had to be passed on to the consumer. The price-sensitive consumers reacted adversely to these price increases and resorted to reducing or deferring consumption. This resulted in a market de-growth of

D-size batteries (the more expensive among various sizes of batteries), which had to bear the highest impact of the cost push, Kumaraswami says.

Torches now use LED bulbs, which consume only one-third of the power consumed by conventional torches using incandescent bulbs. This change in technology is also resulting in lower off-take of D-size batteries.

There is, however, an increase in demand for AA-/AAA-size batteries due to miniaturisation of transistors, increased use of consumer electronic products such as cameras, electronic toys, remotes and quartz clocks, and more importantly growth of newer devices like new-generation LED flashlights.

Technology. In India, carbon-zinc batteries have maintained leading share in the primary (non-rechargeable) battery market for decades due to their lower production costs. The demand for alkaline batteries is forecast to grow in the global market on account of their better performance and a closing price gap with carbon-zinc batteries, particularly in AA and AAA varieties. However, the Indian market will continue to be driven by lower-priced carbon-zinc batteries.

Opportunities. Consumption of batteries is driven by growth in the off-take of end-use appliances. A growing need for portable power and the advent of a number of battery-operated gadgets catalyse consumption. Since these address everyday use, batteries have enjoyed non-cyclical demand.

India has a low per capita consumption across a number of product

groups, batteries included, indicating an inherent potential for growth. Since dry cell batteries are the cheapest source of power, consumption is expected to increase over time. Besides, growing income levels, changing lifestyles and increased need for convenience have resulted in proliferation of battery-powered gadgets such as remote controls, torches, toys, clocks, cameras, FM radio sets and portable music systems, Kumaraswami informs.

Risks. Carbon-zinc batteries do not face any serious threat from alkaline and rechargeable batteries because these are items of recurring use, providing portable energy at an affordable cost. Organised players are adequately protected against competition from imports due to their brand equity, quality and extensive distribution setup.

In any case, all the three organised players have presence in alkaline and rechargeable segments and may be able to participate if the market provides indication of an opportunity.

A tangible threat to battery consumption lies in making the product unaffordable to poorer segment of the economy as seen in the recent past. Weakening of the rupee continues to put pressure on raw material prices. Also, some of the raw material prices are showing signs of hardening. Apart from that, the overall inflationary trends have also been putting pressures on operating costs. These are areas of concern. Fortunately, the market is used to an inflationary environment, where all producers have to pass on adverse impacts faced by them.

LI-ION BATTERIES

'Dry cell' portable rechargeable batteries (in the order of increasing power density and cost) include nickel-cadmium (NiCd), nickel-zinc (NiZn), nickel metal hydride (NiMH) and lithium-ion (Li-ion) cells. Li-ion accounts for the largest share of the dry cell rechargeable market. NiMH has replaced NiCd in most applications due to its higher capacity, but NiCd continues to be used in power tools, two-way radios and medical equipment. NiZn is a new entrant on the block.

Mobile gadgets like cellphones and